Applications/Information

LANL Co-Design Summer School 2025

Model implosion dynamics at scale using FleCSI-HARD

Figure Credit: HARD CDSS24

Model various implosion dynamics and simulate the instabilities associated with these phenomena at scale on Venado:

- Base code is HARD, open source, developed by the CDSS24 students
- Augment the code to model implosion: complex radiation-hydrodynamics scheme with different closures, set of solvers via FleCSolve, variable opacity, and verification + validation
- Optimize and profile the code at scale on the Venado supercomputer (Grace-Hopper architecture) to target large instances of inertial confinement fusion like problems

The School:

- 10 weeks around mid-May to mid-August
- 6 students with multidisciplinary backgrounds (Not restricted to U.S. citizens)
- Working in a Co-Design manner on a common subject

Salary:

$9-12k based on education and experience

Research areas:

- Computer Science (HPC)
- Rad-Hydro, Shock Physics
- Iterative/Multigrid Solvers

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Send any questions to cdss@lanl.gov

LA-UR-24-31774